Lecture 12

20.11.2025

Let us again consider $\quad -u''(x) = f(x) \quad$ in $\Omega = (0,1) \qquad (*)$

$$u(0) = 0 = u(1)$$

Instead of looking for $u \in C^2(\Omega) \cap C(\bar{\Omega})$ satisfying $(*)$, let us consider:

$$\boxed{\text{Find } u \in C_0^1(\bar{\Omega}) \text{ s.t. } \int_\Omega u'(x)v'(x)\,dx = \int_\Omega f(x)v(x)\,dx \text{ for all } v \in C_0^1(\bar{\Omega})}$$

We will refer to this problem (formulation) as variational problem or weak problem

Questions: 1) What is the benefit of this formulation over the one in $(*)$?

2) (When) does this problem have the same solution as $(*)$?

Regarding 1) $\rightsquigarrow$ we do not require u'' to exist, but only u'!

Regarding 2) assume that u solves the variational problem and additionally

$$u \in C_0^2(\bar{\Omega}) := \{ w \in C^2(\bar{\Omega}) \mid w(0) = 0 = w(1) \}$$

$$\Rightarrow \int_\Omega f(x)\,v(x)\,dx = \int_\Omega u'(x)\,v'(x)\,dx \overset{\text{i.b.p.}}{=} -\int_\Omega u''(x)\,v(x)\,dx + \underbrace{[u'(x)\,v(x)]_0^1}_{\text{vanishing}}$$

$$\Rightarrow \int_\Omega (-u''(x) - f(x))\,v(x)\,dx = 0 \quad \text{for } \underline{\text{all}}\ v \in C_0^1(\bar{\Omega})$$

$$\left(\text{means: } \langle -u'' - f, v \rangle_{L^2(\Omega)} = 0 \quad \forall\, v \in C_0^1(\bar{\Omega})\ !\right)$$

Does this imply $-u''(x) - f(x) = 0$? (Spoiler: yes...)

For answering this (positively) we need

1) smooth functions with compact support

2) the Fundamental Lemma of Calculus of Variations

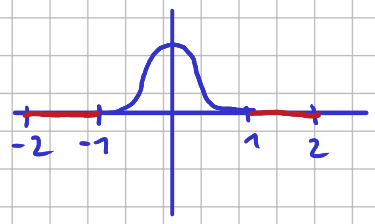
What is this?

1) $C_c^\infty(a,b) := \{ h : (a,b) \to \mathbb{R} \mid h$ is inf. often differentiable and $\operatorname{supp}(h) \subset (a,b)$ is compact $\}$

$$\operatorname{supp}(h) = \overline{\{ x \in (a,b) \mid h(x) \neq 0 \}}$$

How do such functions look like?

For ex.: $h(x) = \begin{cases} \exp\left(-\frac{1}{1-x^2}\right), & x \in (-1,1) \\ 0 & \text{otherwise} \end{cases}$



2) <u>Fundamental Lemma of Calculus of Variations</u> Let $(a,b) \subset \mathbb{R}$ be an open interval and let $g:(a,b) \to \mathbb{R}$ be continuous. IF $\int_a^b g(x)\, h(x)\, dx = 0$ for all $h \in C_c^\infty(a,b)$, then g is identically zero.

$\rightsquigarrow$ We can now apply this lemma to our variational problem

We know: $\int_0^1 \underbrace{(-u''(x) - f(x))}_{=:\, g(x)}\, v(x)\, dx = 0$ for all $v \in C_0^1(\bar{\Omega})$ $(*)$

Note that for a function $h \in C_c^\infty(0,1)$ it holds that

$\operatorname{supp}(h) = \overline{\{x \in (0,1) \mid h(x) \neq 0\}}$ is compact, therefore <u>closed</u>!

$\Rightarrow$ h has to be zero at the boundary points, i.e., $h(0) = 0 = h(1)$

$\Rightarrow$ if $h \in C_c^\infty(0,1)$, then in particular $h \in C_0^1(\bar{\Omega})$

From $(*)$ we therefore conclude that

$$\int_0^1 g(x)\, h(x)\, dx = 0 \quad \text{for all } h \in C_c^\infty(0,1) \subset C_0^1(\bar{\Omega})$$

LEMMA

$\Rightarrow$ $g(x) = 0 \Rightarrow -u''(x) = f(x)$ in $(0,1)$

<u>Conclusion:</u> if u solves the variational problem and additionally $u \in C_0^2(\bar{\Omega})$, then u also solves the original problem

The weak / variational problem can also be interpreted as an <u>optimization</u> problem:

Find $u_{min} \in C_0^1(\bar{\Omega})$ such that $\mathcal{J}(u_{min}) \leq \mathcal{J}(u)$ for all $u \in C_0^1(\bar{\Omega})$

where $\mathcal{J}(u) = \int_\Omega \frac{1}{2}(u'(x))^2 - f(x)\, u(x)\, dx$

Assume $u_{min} \in C_0^1(\bar{\Omega})$ exists and solves this optimization problem.

Consider then an <u>arbitrary</u> but <u>fixed</u> $v \in C_0^1(\bar{\Omega})$ and <u>arbitrary</u> $\alpha \in \mathbb{R}$

$\Rightarrow$ $w := u_{min} + \alpha v \in C_0^1(\bar{\Omega})$ for any $\alpha \in \mathbb{R}$!

u_{min} is minimal $\Rightarrow$ $\underset{\substack{\| \\ j(0)}}{J(u_{min})} \leq J(w) = J(u_{min} + \alpha v) =: j(\alpha)$ for all $\alpha \in \mathbb{R}$!

In other words, the function $j: \mathbb{R} \to \mathbb{R}$, $j(\alpha) = J(u_{min} + \alpha v)$ has a global minimum in $\alpha = 0$! $\Rightarrow$ $j'(0) = 0$

Note: $j'(\alpha) = \frac{d}{d\alpha} \int_\Omega \frac{1}{2} (u_{min}'(x) + \alpha v(x))^2 - f(x)(u_{min}(x) + \alpha v(x))\, dx$

$$= \int_\Omega (u_{min}'(x) + \alpha v(x)) \cdot v(x) - f(x) \cdot v(x)\, dx$$

$$\Rightarrow 0 = j'(0) = \int_\Omega (u_{min}'(x) - f(x))\, v(x)\, dx$$

$$\Leftrightarrow \quad \int_\Omega u_{min}'(x)\, v(x)\, dx = \int_\Omega f(x)\, v(x)\, dx$$

Since $v \in C_0^1(\bar{\Omega})$ was arbitrary, we can do this minimization for any v and obtain

$$\int_\Omega u_{min}'(x)\, v(x)\, dx = \int_\Omega f(x) v(x)\, dx \quad \forall\, v \in C_0^1(\bar{\Omega})$$

$\Rightarrow$ this is again the weak/variational formulation for u_{min}!

(How) does it work in 2D? Consider finding $u: \Omega \to \mathbb{R}$, $\Omega \subset \mathbb{R}^2$ s.t.

$$(*) \quad \begin{aligned} -\Delta u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \Gamma_1 && \text{(homogeneous Dirichlet boundary conditions)} \\ \frac{\partial u}{\partial \nu} &= 0 \text{ on } \Gamma_2 && \text{(homogeneous Neumann boundary conditions)} \end{aligned}$$

where $\Gamma_1 \cup \Gamma_2 = \Gamma = \partial\Omega$ with $\Gamma_1 \cap \Gamma_2 = \emptyset$.

We consider two (generally) different spaces:

- trial space, for this concrete example $V_0 := \{w \in C^1(\bar{\Omega}) \mid w(x) = 0 \text{ on } \Gamma_1\}$
  ↳ here we look for our approximate solution

- test space, for this concrete example also V_0
  ↳ from here we take functions for multiplying ("testing") the PDE

For v in testspace we obtain from $(*)$:

$$-\int_\Omega \Delta u \cdot v\, dx = \int_\Omega f \cdot v\, dx \quad \forall\, v \in V_0$$

Recall multivariate product rule: $\operatorname{div}(v\nabla u) = \nabla u \cdot \nabla v + v \underbrace{\operatorname{div}(\nabla u)}_{=\Delta u}$

$$\Rightarrow \quad -\int_\Omega \Delta u \cdot v \, dx = \int_\Omega \nabla u \cdot \nabla v \, dx - \int_\Omega \operatorname{div}(v \nabla u) \, dx$$

Gauss thm

$$\int_\Omega \operatorname{div}(w) = \int_\Gamma \nu \cdot w$$

Gauss

$$= \int_\Omega \nabla u \cdot \nabla v \, dx - \int_\Gamma \nu \cdot (v \nabla u) \, dA \qquad \forall v \in V_0$$

$$= \int_\Omega \nabla u \cdot \nabla v \, dx - \int_{\underbrace{\Gamma}_{=\Gamma_1 \cup \Gamma_2}} v \cdot \frac{\partial u}{\partial \nu} \, dA = \int_\Omega \nabla u \cdot \nabla v \, dx - \underbrace{\int_{\Gamma_1} v \cdot \frac{\partial u}{\partial \nu} \, dA}_{\substack{=0 \text{ since} \\ v \in V_0}} - \underbrace{\int_{\Gamma_2} v \cdot \frac{\partial u}{\partial \nu} \, dA}_{\substack{=0 \text{ due} \\ \text{to Neumann} \\ \text{b.c.}}}$$

$$\Rightarrow \quad \boxed{\text{Find } u \in V_0 \text{ s.t. } \int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx \quad \forall v \in V_0}$$